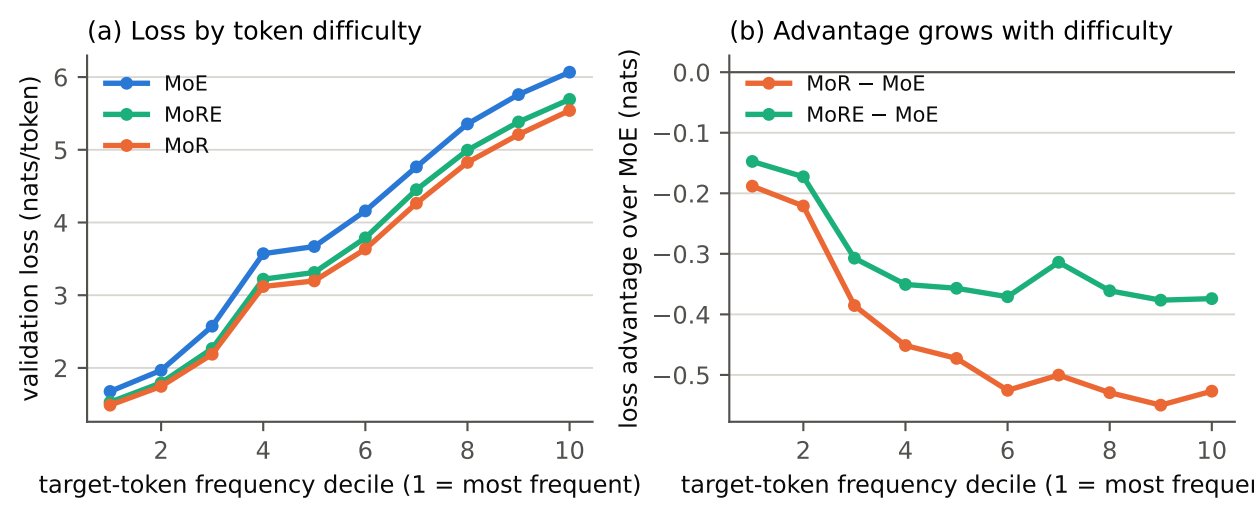
(a) Loss by token difficulty
validation loss (nats/token)
MoE
MoRE
MoR
target-token frequency decile (1 = most frequent)
(b) Advantage grows with difficulty
loss advantage over MoE (nats)
MoR − MoE
MoRE − MoE
target-token frequency decile (1 = most frequen